

## Reparametrization

Def. Let  $\alpha: I \rightarrow \mathbb{R}^3$  be a Curve defined on an open interval  $I$ .

If  $h: J \rightarrow I$  is a differentiable function on an open interval  $J$ , then the composite  $\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$  is a curve called a reparametrization of  $\alpha$  by  $h$ .

Lemma. If  $\beta$  is the reparametrization of  $\alpha$  by  $h$ , then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote  $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ . Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\beta'(s) = (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s))$$

$$= h(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s)))$$

$$= h(s) \cdot \alpha'(h(s)).$$

Thm. Given a regular Curve  $\alpha$  in  $\mathbb{R}^3$ ,

there exists a reparametrization  $\beta$  of  $\alpha$  such that  $\|\beta'(s)\| = 1$  for all  $s$ .

Proof. Let  $\alpha: I \rightarrow \mathbb{R}^3$  be a regular Curve. Let  $a \in I$  be fixed.

Define  $S(t) = \int_a^t \|\alpha'(u)\| du$ . Since  $\|\alpha'\| \neq 0$ , the  $S(t)$  is strictly increasing on  $I$ .

Therefore, there is an inverse function  $t: J \rightarrow I$  s.t.  $S(t(x)) = x$  for all  $x \in J$ , where  $J = S[I]$ . Now, for  $\beta = \alpha \circ t$ ,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \quad \text{and} \quad \|\beta'(s)\| = \underbrace{\frac{dt}{ds}(s)}_{>0} \|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1. \quad \square$$

Note: given a curve  $\alpha$ , define length  $S(t) = \int_a^t \|\alpha'(u)\| du$  and  $\beta = \alpha \circ S^{-1}$ .

Def. Let  $\beta: I \rightarrow \mathbb{R}^3$  be a unit-speed curve.

A vector field  $\beta'$  is called the unit tangent vector field on  $\beta$ , denote  $T = \beta'$ .

The derivative  $T' = \beta''$  is called the curvature vector field of  $\beta$ .

The value  $k(s) = \|T'(s)\|$  ( $s \in I$ ) is called the curvature function of  $\beta$ .

To restriction s.t.  $k > 0$ ,  
the unit-vector field  $N = \frac{T'}{k}$  is called the principal normal vector field of  $\beta$ .

The cross product  $B = T \times N$  on  $\beta$  is called the binormal vector field of  $\beta$ .

Lemma. Let  $\beta$  be a unit-speed curve in  $\mathbb{R}^3$  with  $k > 0$ ,

Then, the three vector fields  $T, N$ , and  $B$  on  $\beta$  are unit vector fields that are mutually orthogonal at each point.

Proof. By definition,  $\|T\| = \|\beta'\| = 1$ . And since the curvature  $k > 0$ ,  $\|N\| = \frac{1}{k} \cdot \|T'\| = 1$ .

And, since  $\|T\|^2 = T \cdot T$  is constant,  $2T \cdot T' = 0$ , so  $T \perp N$ .

therefore the binormal,  $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$ .

In summary,  $T = \beta'$ ,  $N = T'/k$ ,  $B = T \times N$ , and  $\{T, N, B\}$  is orthonormal.

Since  $B \cdot T = 0$ ,  $B' \cdot T + B \cdot T' = 0$  by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot kN = 0$$

And since  $B$  is unit,  $B' \cdot B = 0$ , thus, there is a scalar value function  $\tau$  s.t.

$$B' = -\tau N$$

, called the torsion. Now

Thm. Let  $\beta$  be a unit-speed curve in  $\mathbb{R}^3$  with curvature  $k > 0$  and torsion  $\tau$ ,

$$\begin{aligned} T' &= kN \\ N' &= -kT + \tau B \\ B' &= -\tau N \end{aligned} \quad \begin{bmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe  $N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$

Now,  $N' \cdot T = N \cdot T' = 0$  and  $N' \cdot B = N \cdot B' = 0$  gives the results.

Prop. Let  $\beta$  be a unit-speed curve in  $\mathbb{R}^3$  with  $\kappa > 0$ . Then,

$\beta$  is a plane curve if and only if  $\tau = 0$ , the torsion is zero.

Proof. Suppose that  $\beta$  is a plane curve. That,  $\text{Im } \beta$  lies in some plane.

Therefore, there are two vectors  $p, q \in \mathbb{R}^3$  such that  $q \neq 0$  and

$$[\beta(s) - p] \cdot q = 0 \text{ for all } s.$$

Differentiating yields:

$$\beta'(s) \cdot q = 0 \text{ and } \beta''(s) \cdot q = 0 \text{ for all } s.$$

That is,  $q$  is orthogonal to  $T = \beta'$  and  $N = \beta''/\kappa$ . So,  $\beta$  and  $q$  are parallel.

We can conclude that  $\beta = \pm q/||q||$ . So  $\beta' = 0$ , the torsion  $\tau$  must be zero.

Conversely, suppose that the torsion is zero. Then,  $\beta' = 0$  by definition.

( $\beta' = 0$  means  $\beta$  is parallel.) To show that  $\text{Im } \beta$  lies in some plane, define

$$f(s) = [\beta(s) - \beta(0)] \cdot \beta(s) \text{ for all } s.$$

Then, the derivative is

$$f'(s) = \beta'(s) \cdot \beta(s) + \beta(s) \cdot \beta'(s) \stackrel{\text{def.}}{=} \beta'(s) \cdot \beta(s) = T \cdot \beta = 0.$$

By definition,  $f(0) = 0$ , so  $f = 0$ . Consequently,

$$[\beta(s) - \beta(0)] \cdot \beta = 0 \text{ for all } s,$$

So proved  $\square$

Prop. Let  $\beta$  be a unit speed curve with constant curvature  $\kappa > 0$  and torsion zero.

Then, the  $\beta$  is part of a circle of radius  $1/\kappa$ .

Proof. Since  $\tau = 0$ , the  $\beta$  is a plane curve. Define a curve

$$\gamma(s) = \beta(s) + \frac{1}{\kappa} \cdot N(s).$$

By the Frenet formulae,  $\gamma' = \beta' + \frac{1}{\kappa} \cdot N' = T + \frac{1}{\kappa} \cdot (-\kappa T) = 0$ . Therefore,  $\gamma$  is constant.

Now,  $d(\gamma, \beta(s)) = \|\gamma - \beta(s)\| = \frac{1}{\kappa} \cdot \|N\| = \frac{1}{\kappa}$ , so proved.

# General version of Frenet formula,

Let  $\alpha: I \rightarrow \mathbb{R}^3$  be any regular curve, not necessary  $|\alpha'| \equiv 1$ .

And, let  $\beta: J \rightarrow \mathbb{R}^3$  be a reparametrization of  $\alpha[I]$  by the arc-length

$$s = s(t) = \int_{t_0}^t |\alpha'(u)| du, \quad \text{where } t_0 \in I.$$

With the inverse function  $t = t(s) = s^{-1}$ . Then,

1).  $\frac{dt}{ds} = \frac{1}{|\alpha'|}$  and  $\frac{d^2t}{ds^2} = -\left(\frac{\alpha' \cdot \alpha''}{|\alpha'|^4}\right)$

2). The curvature and torsion of  $\alpha$  is given by

$$K(t) = \frac{|\alpha' \times \alpha''|}{|\alpha'|^3} \quad \text{and} \quad \tau(t) = -\frac{(\alpha' \times \alpha'') \cdot \alpha''}{|\alpha' \times \alpha''|^2}.$$

Proof. Since  $\frac{dt}{ds} = \left(\frac{ds}{dt}\right)^{-1} = \left(\frac{d}{dt} \int_{t_0}^t |\alpha'(u)| du\right)^{-1} = \frac{1}{|\alpha'|}$ , the first is obtained

And,  $\frac{d}{ds} \frac{dt}{ds} = \frac{d}{ds} \left( \frac{1}{|\alpha'(t)|} \right)$

Signed Curvature for plane curve.

Let  $\alpha: I \rightarrow \mathbb{R}^2$  be a plane curve.

Then, the binormal  $B$  is constant vector, so we consider only  $T$  and  $N$ .